

Project 63: Language Identification from Audio

Goal: Create a tool that:

1. Accepts audio input (mic or file)
 2. Analyzes the speech pattern
 3. Returns the **most probable spoken language** (e.g., English, Spanish, Hindi, etc.)
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How It Works

You'll use models like:

- **OpenAI Whisper** (supports automatic language detection)
 - **pycld3** (Google's Compact Language Detector for text)
 - Optional: **VoxLingua107**, **SpeechBrain**, or **FastText** for advanced use
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Step-by-Step Implementation (Using Whisper)

Step 1: Install Required Libraries

```
pip install openai-whisper
```

Step 2: Load the Model and Audio

```
import whisper

# Load Whisper model
model = whisper.load_model("base") # or "small", "medium", etc.

# Transcribe with language detection
result = model.transcribe("sample_audio.mp3", task="transcribe", language=None)

# Detected language
print("🌐 Detected Language:", result["language"])
```

Whisper Result Sample

🌐 Detected Language: es # Spanish

Whisper returns **ISO 639-1 codes** (e.g., en, fr, hi, zh).

Step 3: Convert Language Code to Full Name

```
import pycountry

def get_language_name(lang_code):
    try:
        return pycountry.languages.get(alpha_2=lang_code).name
    except:
        return lang_code

lang_name = get_language_name(result["language"])
print("✅ Language Name:", lang_name)
```

Optional: Use Whisper Just for Detection (No Transcription)

```
audio = whisper.load_audio("sample_audio.mp3")
audio = whisper.pad_or_trim(audio)
mel = whisper.log_mel_spectrogram(audio).to(model.device)

_, probs = model.detect_language(mel)
detected_lang = max(probs, key=probs.get)
print("🔊 Most Likely Language:", get_language_name(detected_lang))
```

Project Summary

- **Input:** Audio file or mic input
- **Output:** Detected spoken language (ISO code + full name)
- **Model:** Whisper (supports multilingual speech recognition)
- **Use Case:** Auto-routing, localization, transcription prep